

COURSE TITLE : **OBJECT ORIENTED WEB APPLICATION FRAMEWORKS**
 COURSE CODE : **6261**
 COURSE CATEGORY : **A**
 PERIODS/WEEK : **5**
 PERIODS/SEMESTER : **75**
 CREDITS : **5**

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Object Oriented PHP basics	18
2	Advanced Object Oriented PHP concepts	19
3	MVC Frameworks and CodeIgniter basics	19
4	CodeIgnater MVC Framework	19

Course General Outcomes:

Sl.	G.O	On completion of this course the student will be able to :
1	1	To understand Object Oriented PHP basics
2	1	To understand Advanced Object Oriented PHP concepts
3	1	To understand MVC Frameworks and CodeIgniter basics
4	1	To understand CodeIgnater MVC Framework

Specific Outcomes:

MODULE – I: Object Oriented PHP basics

1.1 Understand Object Oriented PHP basics

- 1.1.1** Define Object Oriented programming
- 1.1.2** Discuss why object orientation
- 1.1.3** Describe object orientation concepts
- 1.1.4** Discuss history and evolution of Object Oriented Programming
- 1.1.5** Discuss Need of object oriented programming in software industry
- 1.1.6** List its advantages over structured programming
- 1.1.7** Define classes and objects

- 1.1.8 Discuss OOP features like abstraction, Encapsulation, polymorphism, inheritance, Information hiding, runtime polymorphism and overriding functions
- 1.1.9 Explain object properties and behaviour
- 1.1.10 Describe Access specifiers for object
- 1.1.11 Discuss evolution of object Orientation in PHP
- 1.1.12 Differentiate structured and object oriented PHP
- 1.1.13 Discuss how to define classes in PHP
- 1.1.14 Discuss how to define properties and functions
- 1.1.15 Explain Requiring classes into WebPages
- 1.1.16 Explain how to Create objects inside WebPages
- 1.1.17 Explain accessing object functions and properties

MODULE – II: Advanced Object Oriented PHP concepts

2.1 Understand Advanced Object Oriented PHP concepts

- 2.1.1 Define access specifiers
- 2.1.2 Discuss public, private and protected access specifiers
- 2.1.3 Define *\$this* object pointer
- 2.1.4 Discuss why *\$this* is mandatory in PHP
- 2.1.5 Relate *\$this* and polymorphism
- 2.1.6 Create constructors and destructors
- 2.1.7 Explain Inheritance in PHP
- 2.1.8 Differentiate multiple and multilevel inheritance
- 2.1.9 Explain inheritance and Inheriting Constructors
- 2.1.10 Explain Over riding methods
- 2.1.11 Define the scope resolution operator
- 2.1.12 Define static members
- 2.1.13 Describe abstract classes and methods
- 2.1.14 Define traits
- 2.1.15 Explain Interfaces

MODULE – III: MVC Frameworks and CodeIgniter basics

3.1 Understand MVC Frameworks and CodeIgniter basics

- 3.1.1 Define frameworks
- 3.1.2 Discuss need for a framework in developing web Applications
- 3.1.3 Define the MVC framework
- 3.1.4 Explain Architecture and structure of an MVC framework
- 3.1.5 Explain the Controller – Models – Views - Bootstrap scripts
- 3.1.6 List Advantages and features of MVC methodology
- 3.1.7 List MVC framework examples in PHP
- 3.1.8 Discuss Content Managements systems based on MVC
- 3.1.9 Define CodeIgniter framework
- 3.1.10 Explain MVC Architecture of CodeIgniter (2.x) Framework
- 3.1.11 Explain installation of CodeIgniter in LAMP server
- 3.1.12 Describe CodeIgniter initial configurations - the config file
- 3.1.13 Create controller – models – views
- 3.1.14 Develop calling models from controllers and loading views
- 3.1.15 Describe URL management

MODULE – IV: CodeIgnater MVC Framework

4.1 Understand CodeIgnater MVC Framework

- 4.1.1 List CodeIgniter libraries
- 4.1.2 Use loading a library into controllers and models
- 4.1.3 Describe database library
- 4.1.4 Explain Active records in database library
- 4.1.5 Discuss configuring active record
- 4.1.6 Describe database drivers
- 4.1.7 Use Form validation library
- 4.1.8 Explain setting validation rules
- 4.1.9 Describe managing errors and labels
- 4.1.10 Demonstrate repopulating forms
- 4.1.11 Explain session library
- 4.1.12 Show managing user sessions
- 4.1.13 Use file uploading library
- 4.1.14 Define helpers
- 4.1.15 Distinguish loading helpers, form helper, URL helper, cookie helpers
- 4.1.16 Explain Routing configurations

CONTENT DETAILS

MODULE – I: Object Oriented PHP basics

Introduction object Oriented programming – why object orientation – object orientation concepts – history and evolution of Object Oriented Programming – Need of object oriented programming in software industry – advantages over structured programming – classes and objects – features – abstraction – Encapsulation – polymorphism - inheritance – Information hiding – Runtime polymorphism and overriding functions – object properties and behaviour -Access specifiers for object - evolution of object Orientation in PHP – difference between structured and object oriented PHP - Defining classes in PHP – defining properties and functions - Requiring classes into WebPages – creating objects inside WebPages – accessing object functions and properties.

MODULE – II: Advanced Object Oriented PHP concepts

access specifiers – public, private and protected access specifiers - *\$this* object pointer – why *\$this* is mandatory in PHP – Relation between *\$this* and polymorphism – creating constructors and destructors – Inheritance in PHP – Difference between multiple and multilevel inheritance – Implementing inheritance – Inheriting constructors – Over riding methods – the scope resolution operator – static members – Abstract classes and methods – traits – Implementing Interfaces.

MODULE – III: MVC Frameworks and CodeIgniter basics

Introduction to frameworks – What is a framework - need for a framework in developing web Applications – the MVC framework - Architecture and structure of an MVC framework – The controller – Models – Views – Bootstrap scripts – Advantages and features of MVC methodology – MVC framework examples in PHP- Content Managements systems based on MVC - Introduction CodeIgniter framework – MVC Architecture of CodeIgniter (2.x) Framework – Installing CodeIgniter in LAMP server - CodeIgniter initial configurations – the config file – creating a controller – creating models – creating views – calling models from controllers – loading views – URL management.

MODULE – IV: CodeIgnater MVC Framework

CodeIgniter libraries – loading a library into controllers and models – database library – Active records in database library – configuring active record – database drivers - Form validation library – setting validation rules – managing errors and labels - repopulating forms - session library – managing user sessions – file uploading library - helpers – loading helpers – form helper – URL helper – cookie helpers - Routing configurations.

TEXT BOOK(S)

1. PHP: The Complete Reference by Steven Holzner , McGraw Hill Education.
2. CodeIgniter 2 Cookbook, by Rob Foster

REFERENCE

1. HTML 5 Black Book: Covers CSS3, Javascript, XML, XHTML, AJAX, PHP and jQuery. By Kogent Learning Solutions Inc